Utrecht University

Multi-Criteria Decision Analysis (very brief) Introduction

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Content

Why

What

How

Motivation

Why are environmental decisions difficult?

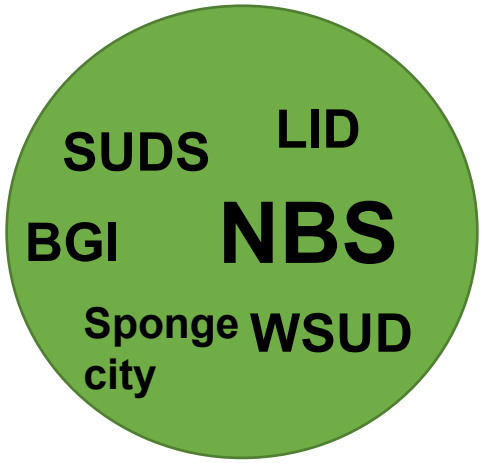
Adapted from Judit Lienert

Motivation

Why are environmental decisions difficult?

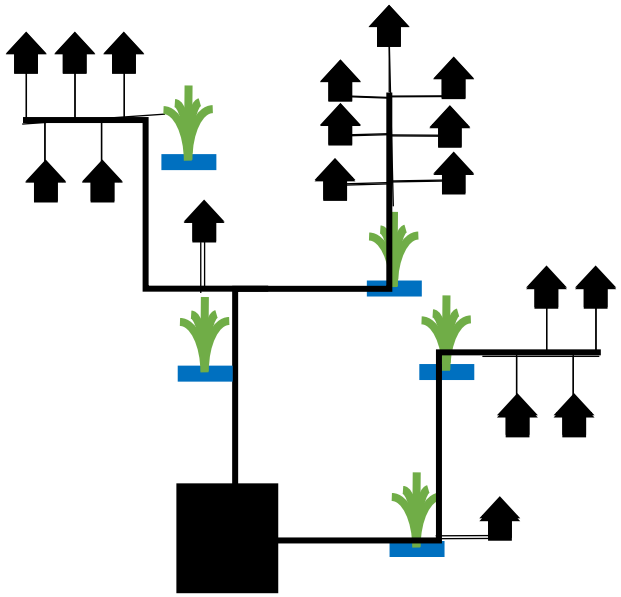
- Large number of alternatives
- Potentially conflicting objectives
- Large uncertainty
  * of data
  * of consequences of alternatives
  * of the future
- Interdisciplinary nature of problem
- Many stakeholders involved
- ...

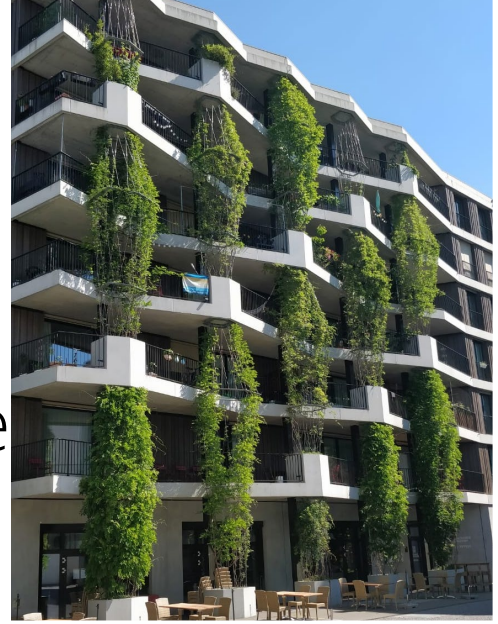
Adapted from Judit Lienert

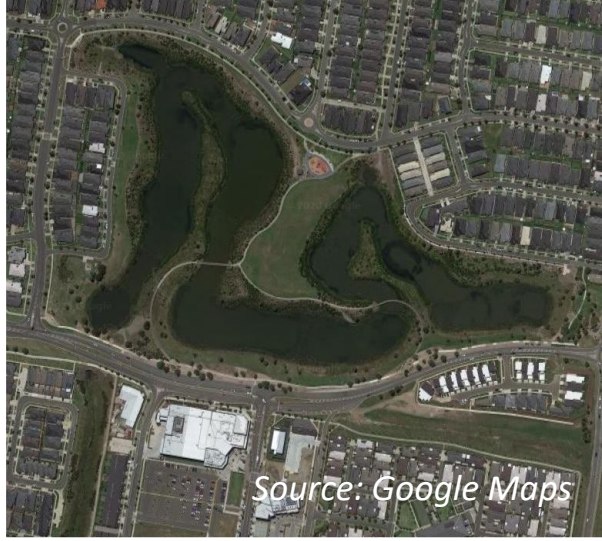


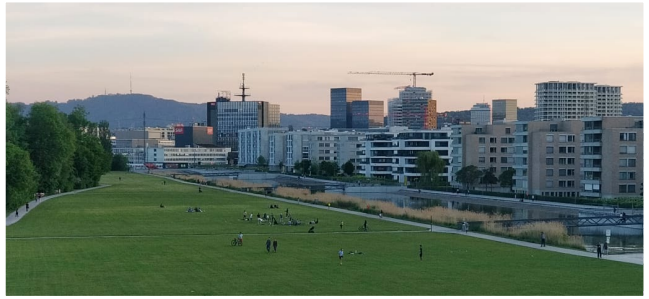
The elements

- Ponds/Lakes
- Green infrastructure
- Rain tanks
- Local sewer









Source all images unless indicated otherwise: Martijn Kuller

Goal of decision analysis

Support the decision process to make better decisions

Definition

"... decision analysis is a formalization of common sense for decision problems which are too complex for informal use of common sense."

"... a philosophy, articulated by a set of logical axioms and a methodology and collection of systematic procedures (...) for responsibly analyzing the complexities inherent in decision problems."

Ralph L. Keeney (1982) Oper. Res. 30: 803-838

Adapted from Judit Lienert

What is a good decision?

- Careful consideration of objectives
- Use of scientific information about consequences of *all* alternatives
- Participatory and transparent
- Considers values of those involved
- ≠ Lucky decision
- Does not always lead to optimal consequences (limited knowledge, chance)

→ Decision support does *not* make the decision for the decision maker

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Adapted from Judit Lienert

Poll

What can we do?

What is important to me?

Poll

Alternatives

Objectives

Content

Why

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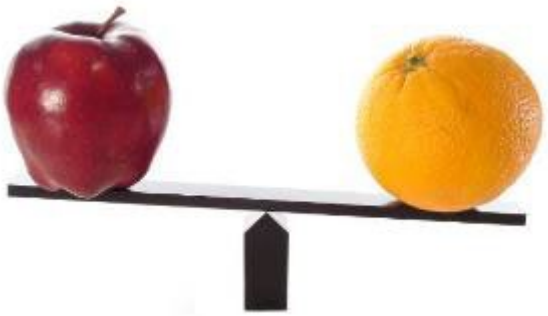
MCDA – the Method

Concept of Multi-Criteria Decision Analysis (MCDA)

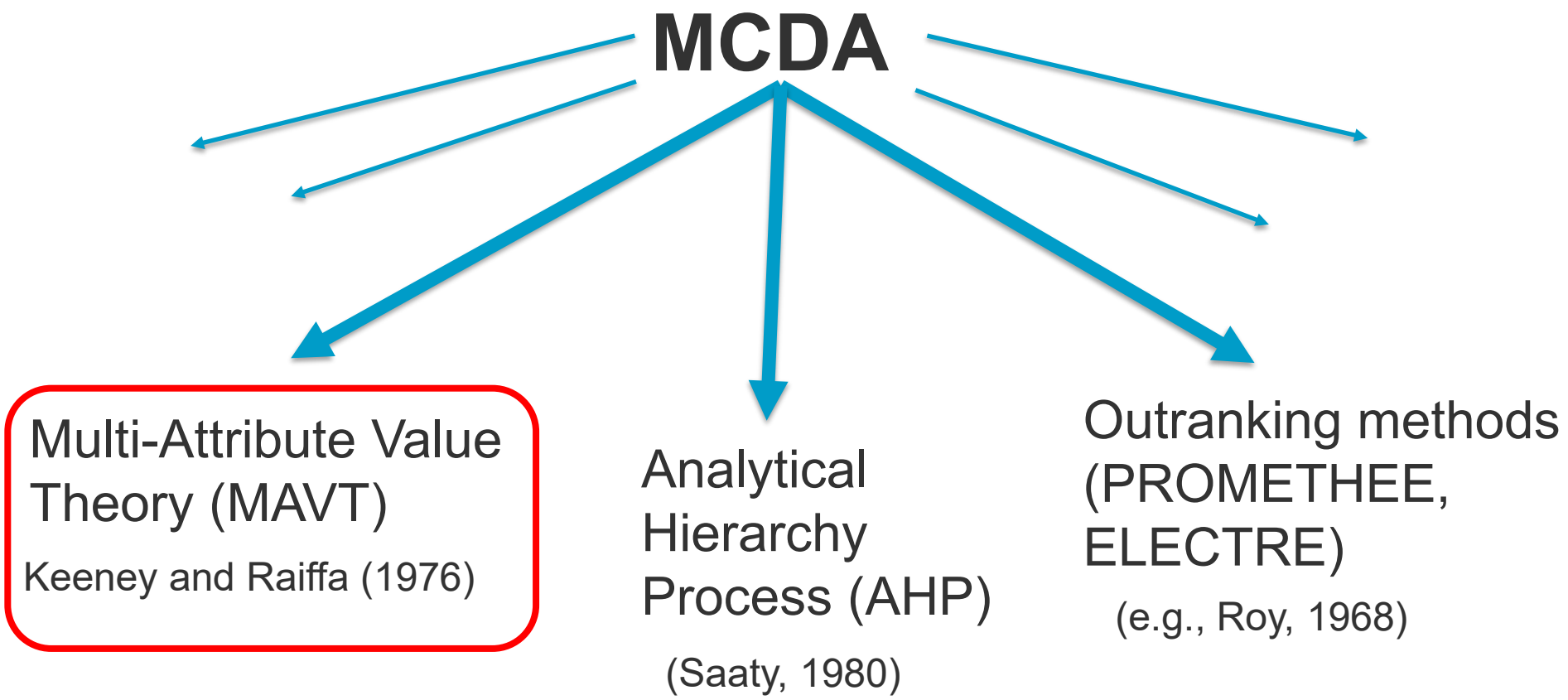
How to compare apples with oranges?



How to deal with a whole fruit basket?!

Adapted from Judit Lienert

Schools of MCDA



Main concepts

Alternatives

The different decision options

- Grey infrastructure
- Blue-Green infrastructure
- Centralised system
- Decentralised system

- Wetlands
- Rain gardens
- Green roofs
- Policies
- Subsidies
- Etc… Where?

Attributes

Measures the performance of an alternative on an objective

- Death rate (Deaths/cap./an.)
- Total damage ($/cap./an.)
- Biodiversity (Species/m^2)
- Water savings (L/cap./an.)

Objectives

Describes what is really important to you

- Good public health
- Low damage/disruption
- Good env. health
- Low costs

- Mitigate flooding
- Provide cooling
- Improve biodiversity
- Beautification
- Reduce water usage
- Etc.

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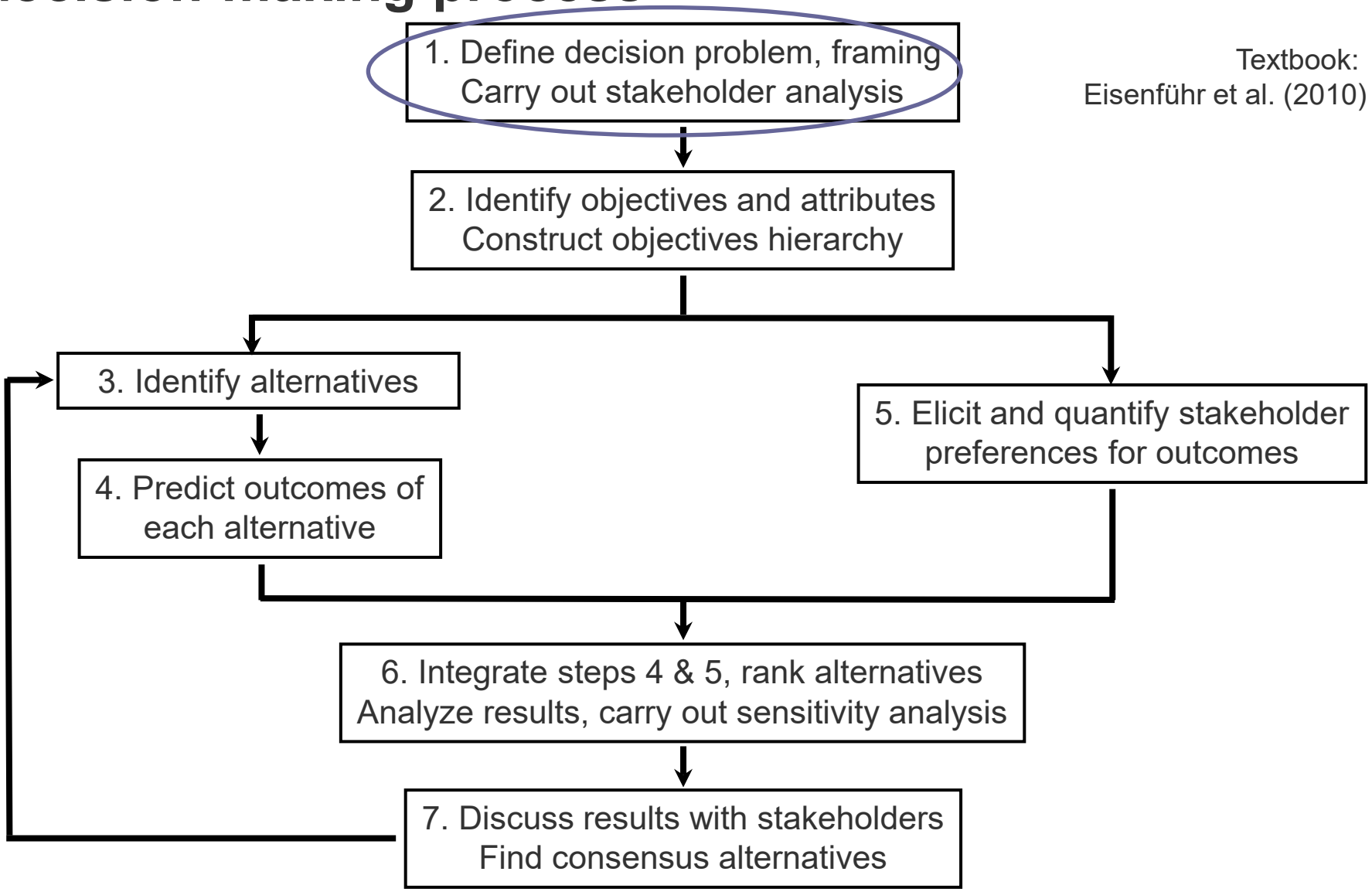
MCDA – the Method: decision-making process

Textbook:
Eisenführ et al. (2010)



Adapted from Judit Lienert

MCDA – 1. Structure decision

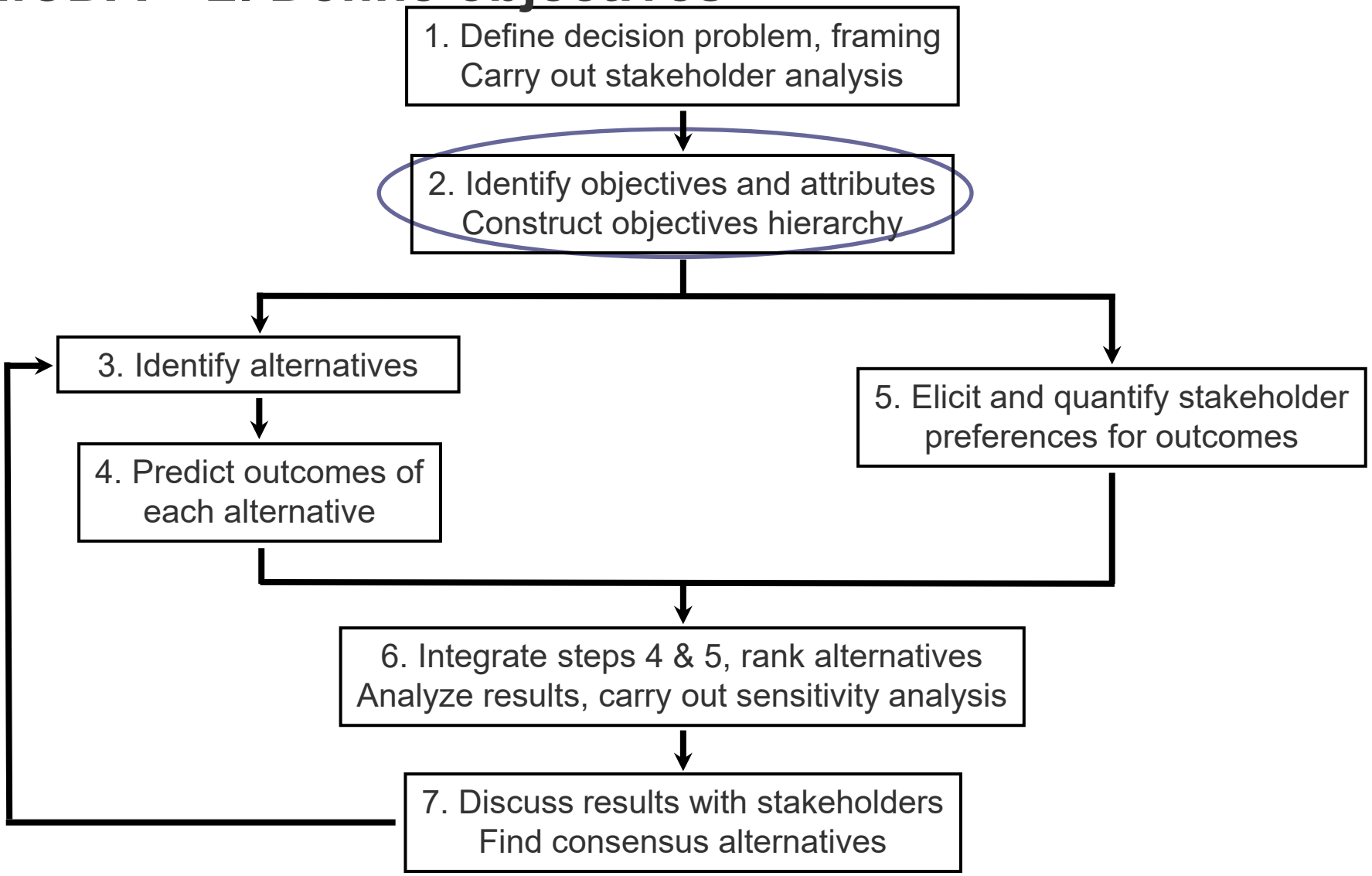
- Aim of the decision making process?
- What to include? What to exclude? System boundaries?
- Who is affected by the decision? Who makes the decision? What are their interests? Whom to include in decision process?
- Where do you get the information? (documents, regulations, experts, models, …)
- Your resources? (money, time, …)
- …

What exactly?

Who decides?
Who is affected?

Adapted from Judit Lienert

MCDA – 2. Define objectives



Adapted from Judit Lienert

MCDA – 2. Define objectives

What is *really* important (to you)?

In *this* decision!!

How to *measure* how well objective is achieved?

(attributes = indicators)

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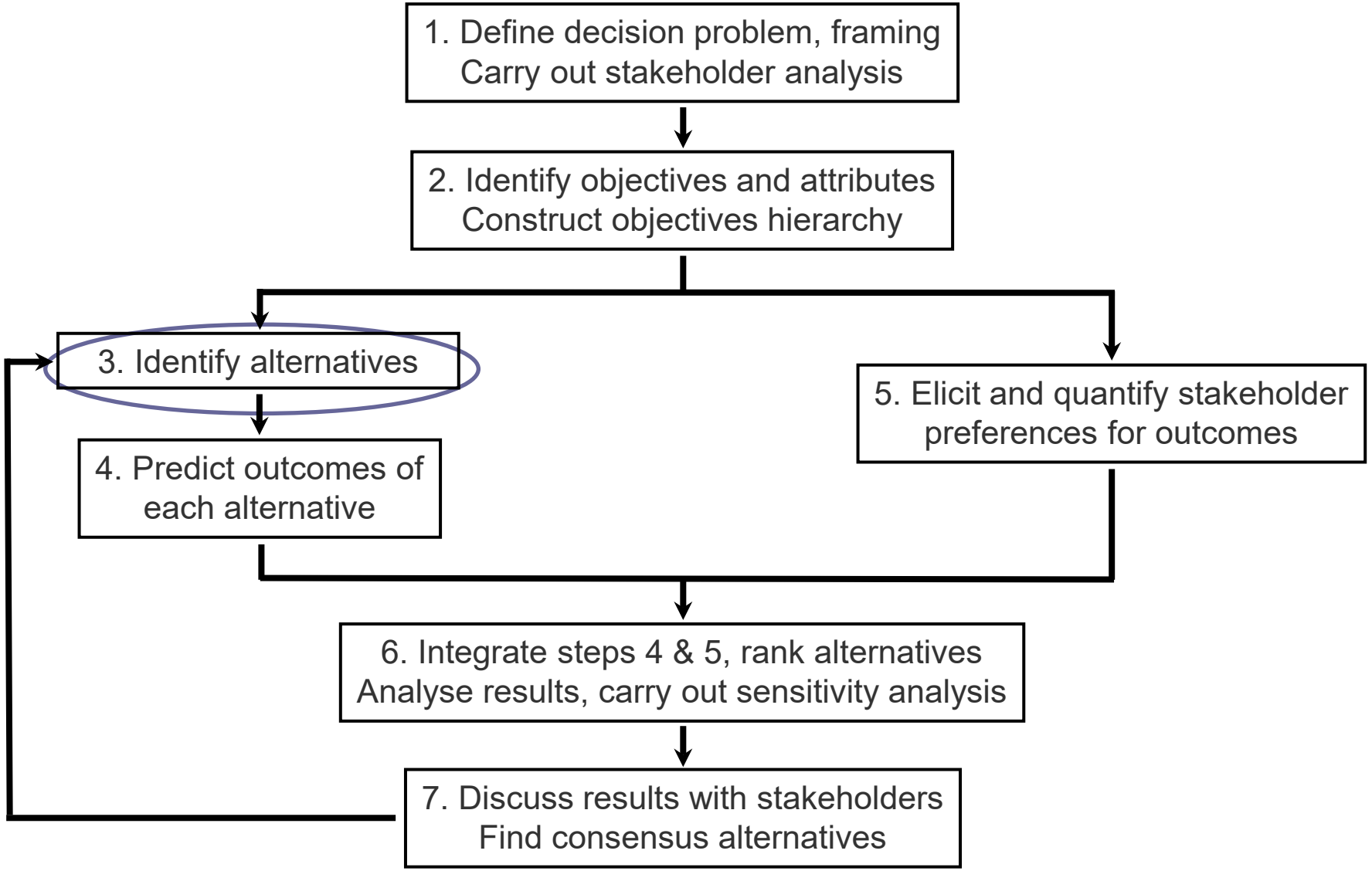
- Focus on fundamental objectives: this is important *«because»* it is important
- Avoid means objectives: important to achieve a more fundamental objective
- Objectives should not contain «means-ends relationships» (objective A ↔ B)
- Complete, Non-redundant, Measurable, Preferentially independent, Simple, Concise, With direction

Adapted from Judit Lienert

MCDA – 3. Identify alternatives (options)



Adapted from Judit Lienert

MCDA – 3. Identify alternatives (options)

How can we achieve the objectives?

(alternatives, options, variants, systems, measures, …)

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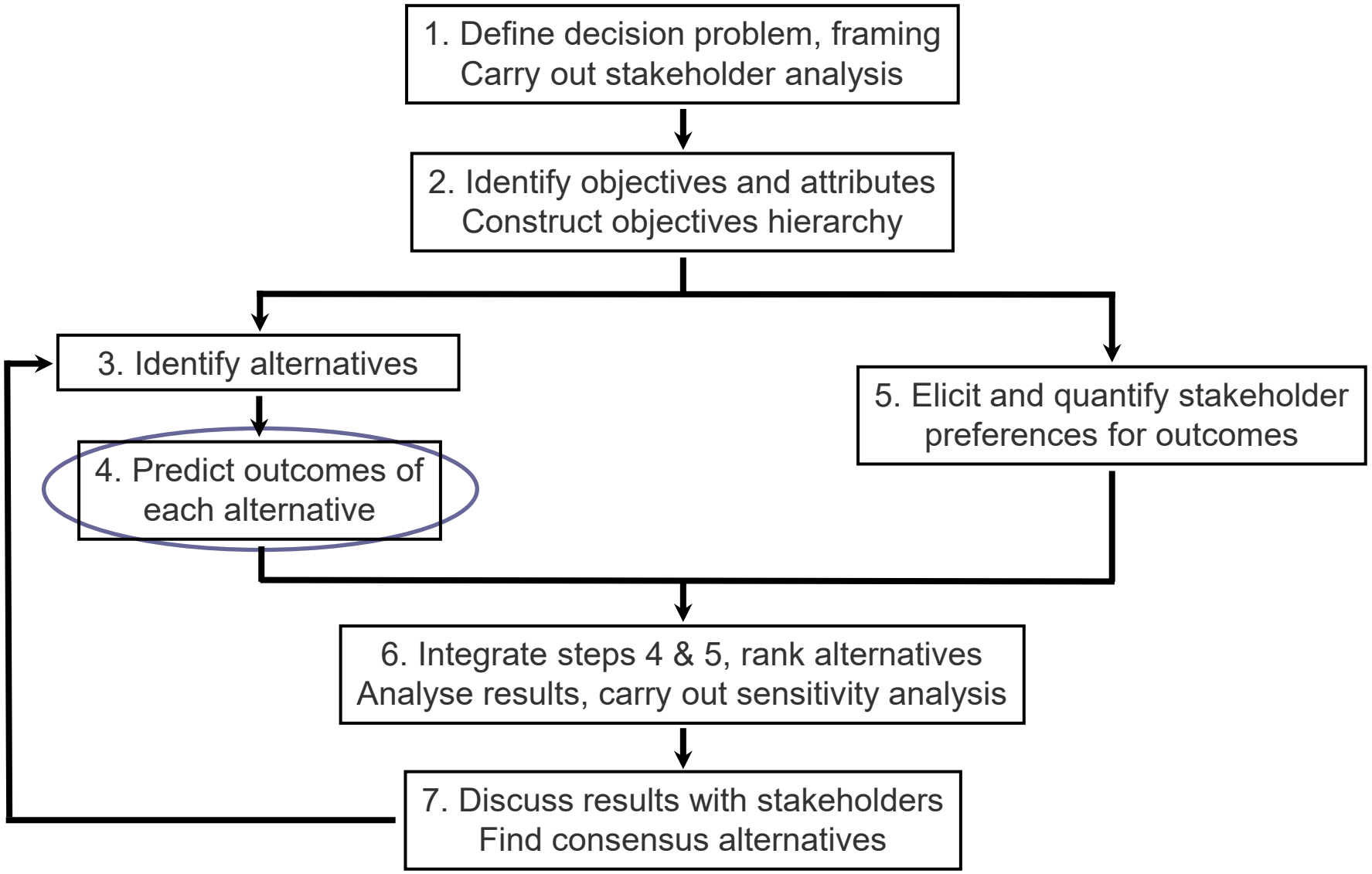
- Think outside the box (no status-quo)
- Use systematic creativity methods
- Make alternatives diverse

Adapted from Judit Lienert

MCDA – 4. Make predictions



Adapted from Judit Lienert

MCDA – 4. Make predictions

For each alternative: How well are objectives achieved?

How large is uncertainty of this estimate?

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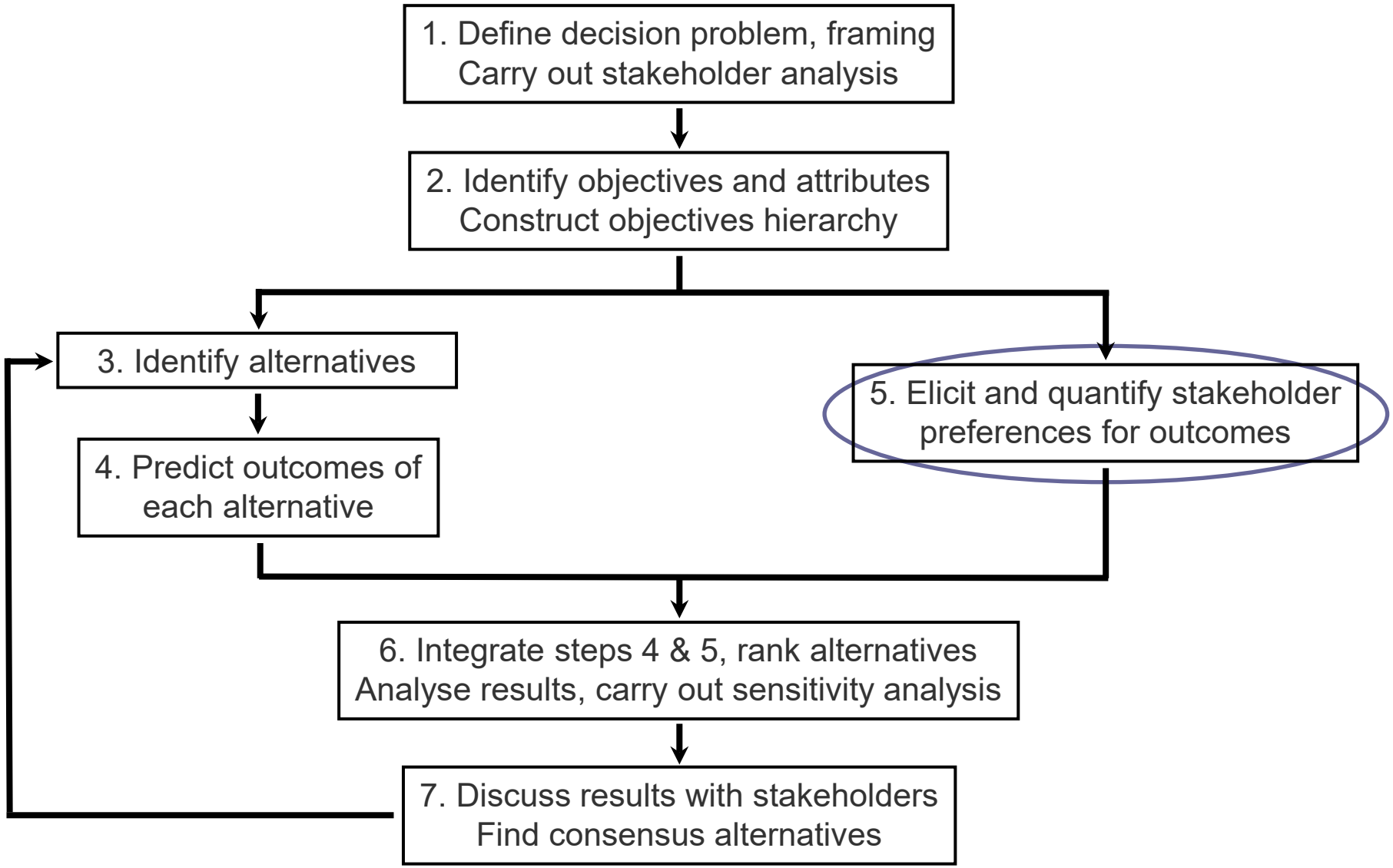
- Scientific predictions («objective» predictions, estimates)
- Based on models, data from literature, expert estimates
- Can be very detailed or coarse, depending on decision problem

Adapted from Judit Lienert

MCDA – 5. Elicit stakeholder preferences



Adapted from Judit Lienert

MCDA – 5. Elicit stakeholder preferences

How important is objective compared to the others?

Which trade-offs are you (she, he..) willing to make?

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- Translate attributes into a common scale (value functions)

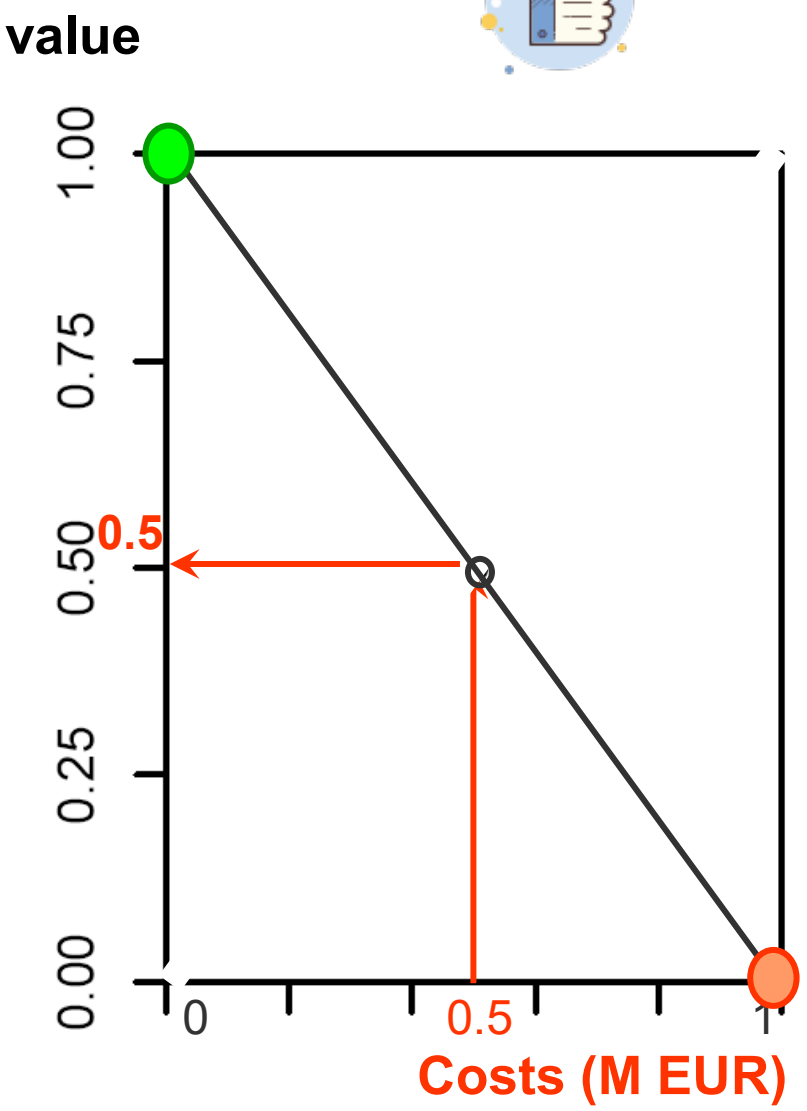
- Compare importance objectives to each other (weights)

Adapted from Judit Lienert

MCDA – 5. Elicit stakeholder preferences

Single-attribute value functions

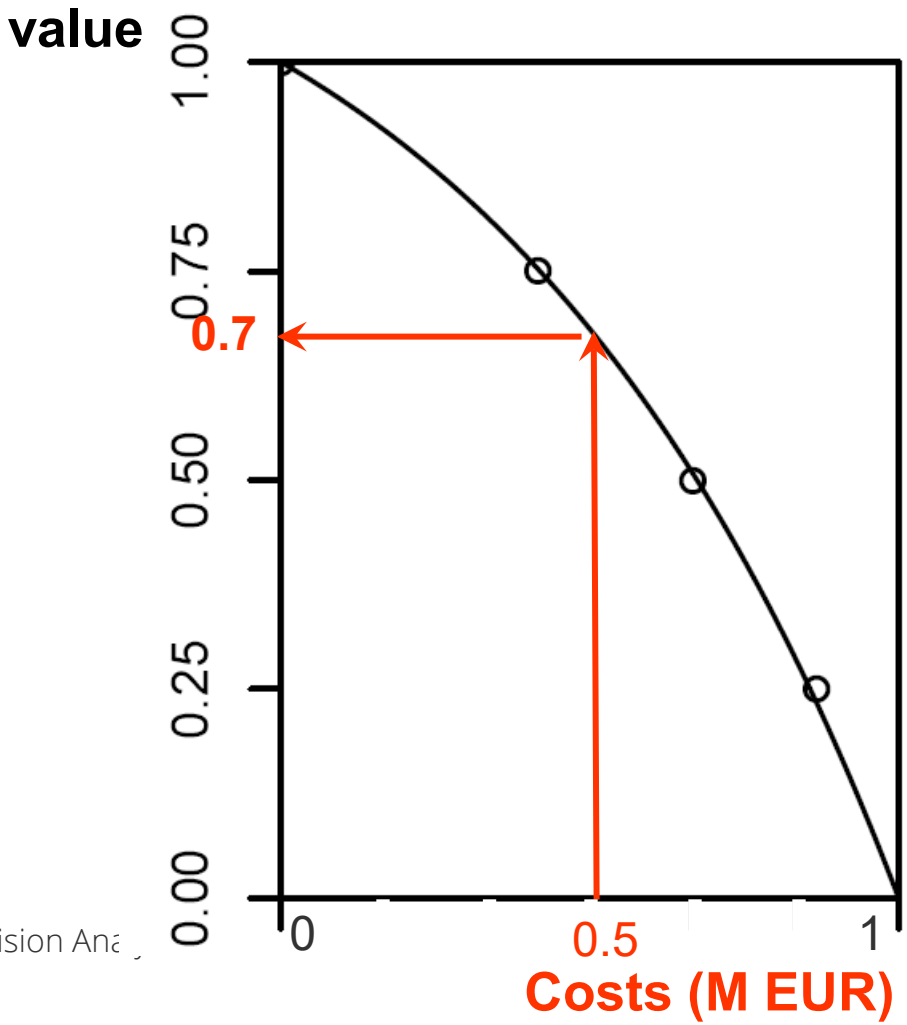
- Quantifies the degree of fulfillment of each attribute
- Common unit between 0 and 1 by transforming level of an attribute to a value
- 0 = This objective is not achieved, 1 = fully achieved
- Single-attribute value functions thus allow to compare attributes with different units
- Can have any shape
- Caveat! Attribute range!



Adapted from Judit Lienert

MCDA – 5. Elicit stakeholder preferences

Single-attribute value functions



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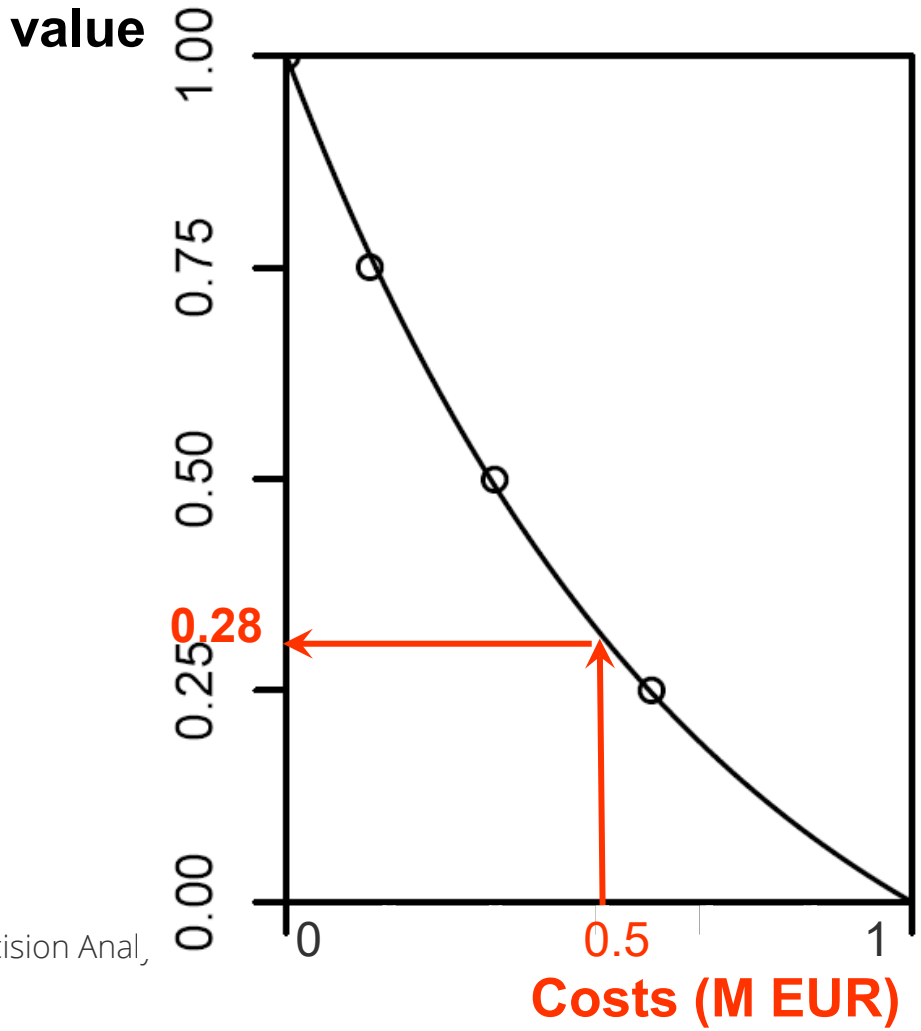
MCDA – 5. Elicit stakeholder preferences

Single-attribute value functions



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MCDA – 5. Elicit stakeholder preferences

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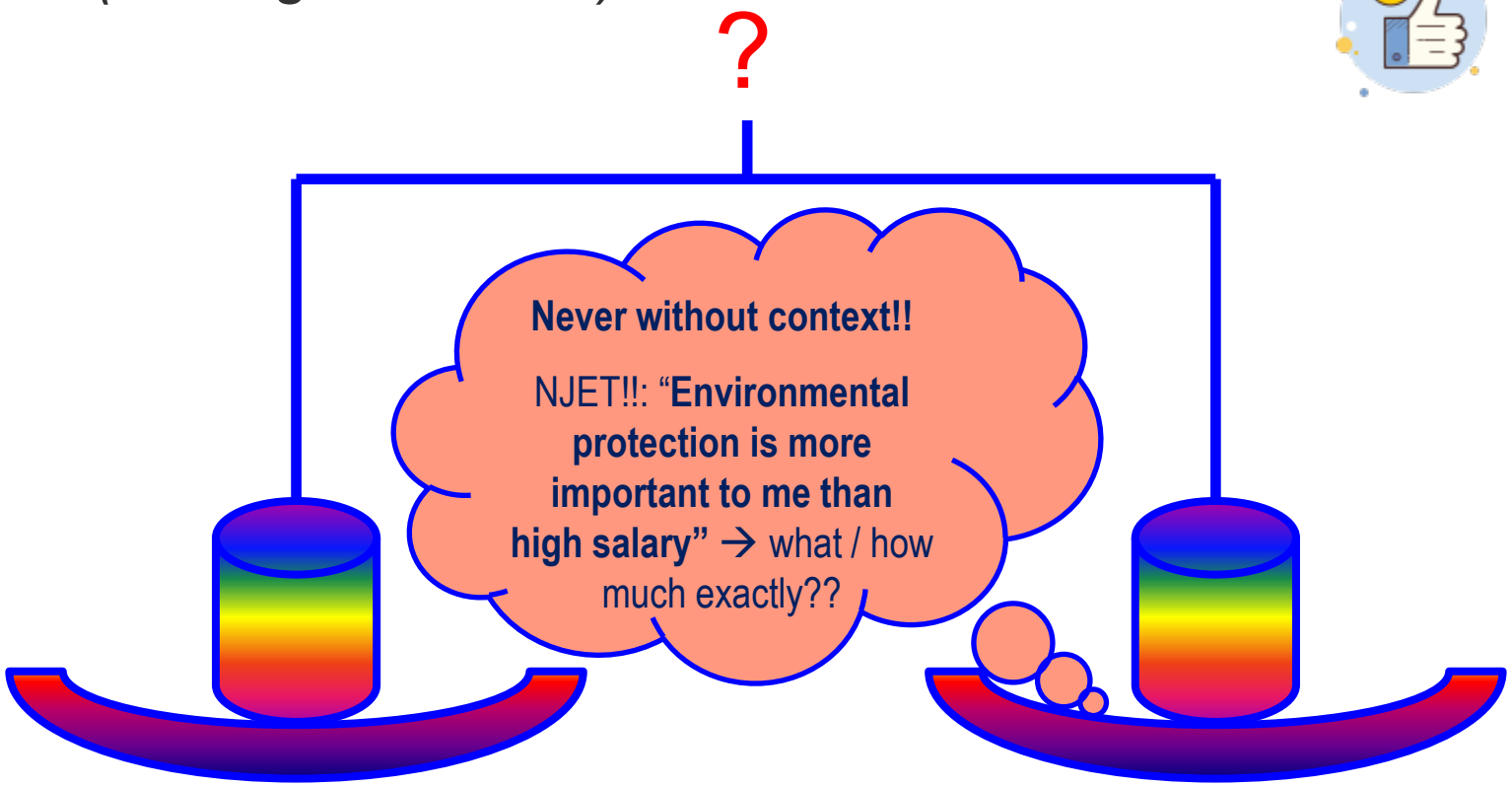
MCDA – 5. Elicit stakeholder preferences

Weights (scaling constants)



Low cost
(0 – 1 M EUR/year)

High nutrient removal
(0 – 81 % P-removal)

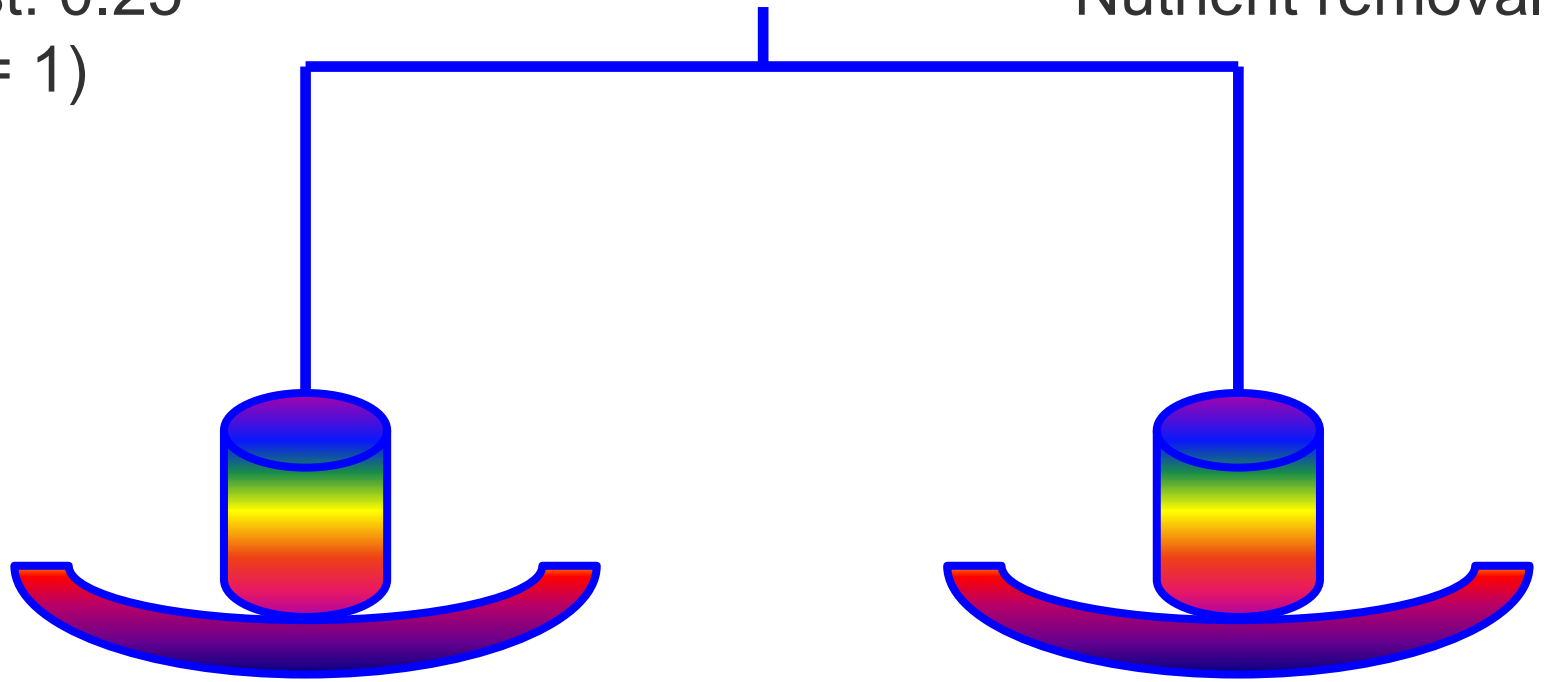
Adapted from Judit Lienert

MCDA – 5. Elicit stakeholder preferences

Weights (scaling constants)

Cost: 0.25
(∑ = 1)

Nutrient removal: 0.75



Low cost (0 – 1 M EUR/year)

High nutrient removal
(0 – 81 % P-removal)

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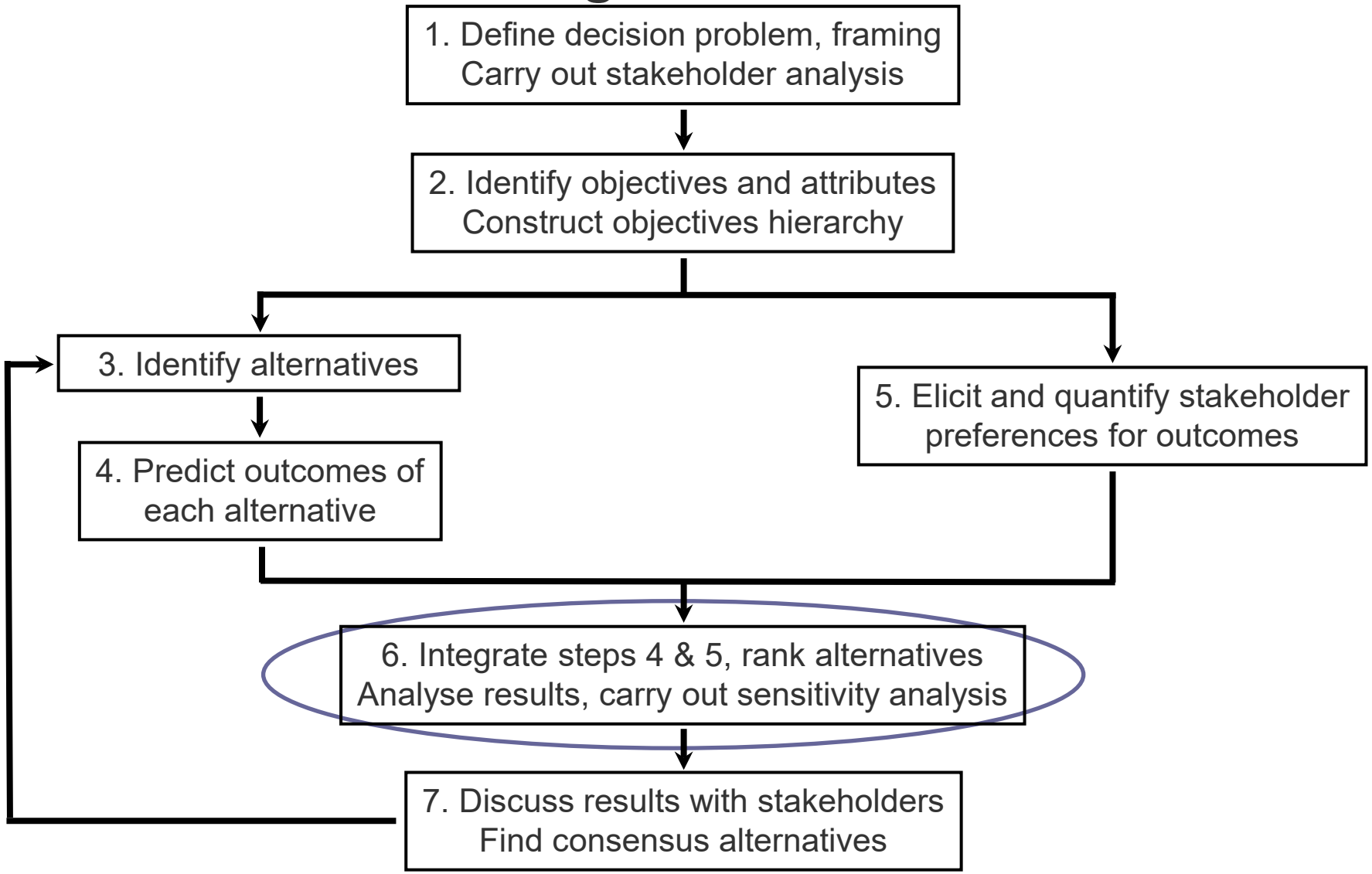
MCDA – 6. Model = Integration



Adapted from Judit Lienert

MCDA – 6. Model = Integration

How well are objectives achieved by each alternative

(and given the preferences of each stakeholder)?

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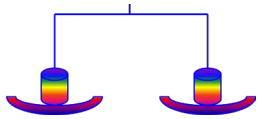
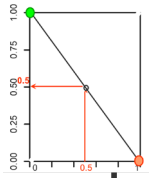
Alternative performance =



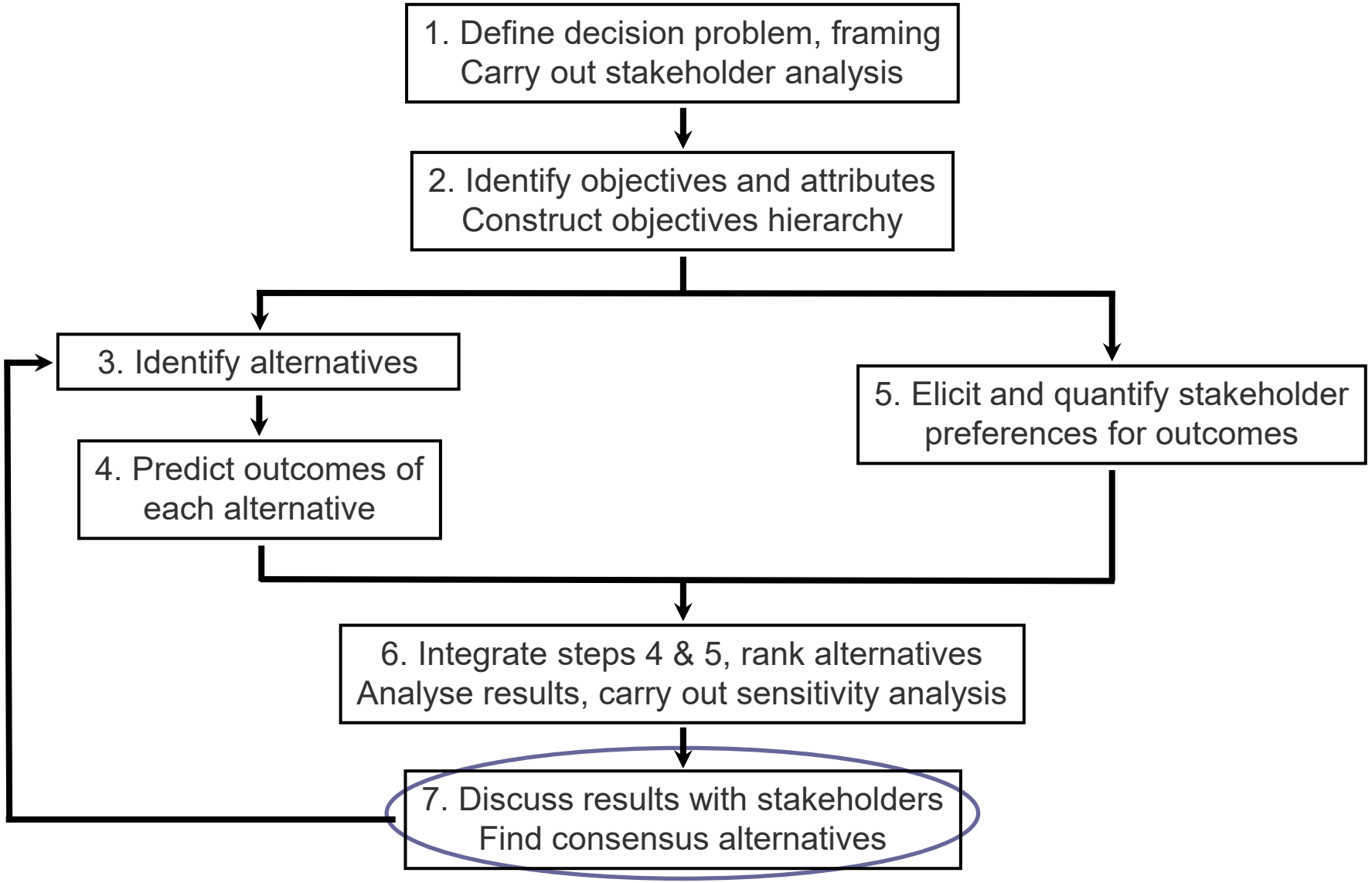


weight objective 1 x achieved value +
weight objective 2 x achieved value +
… +
weight objective *n* x achieved value)

0 ≤ alternative performance ≤ 1

Adapted from Judit Lienert

MCDA – 7. Discuss results with stakeholders



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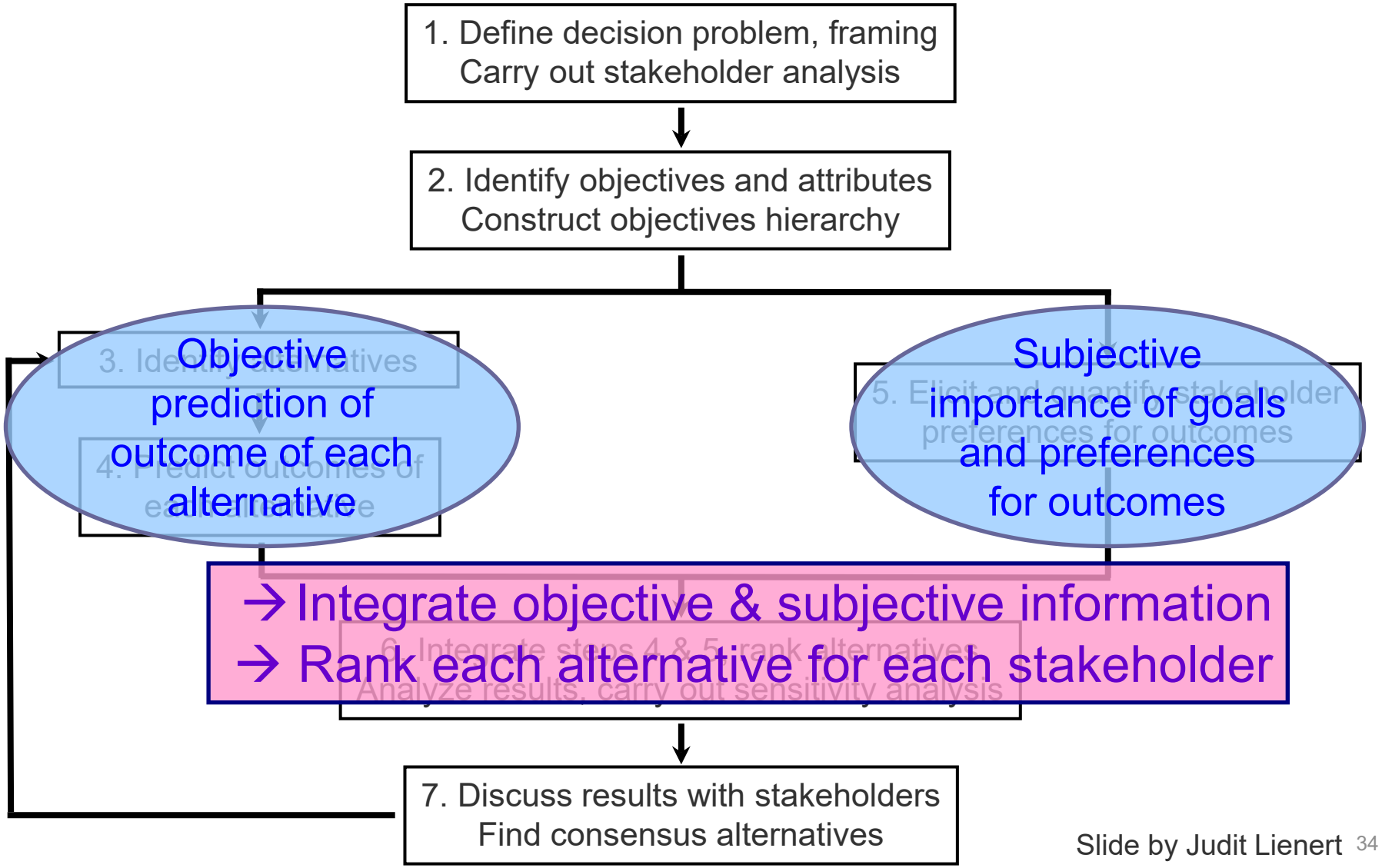
MCDA – the Method: decision-making process



GIS-MCDA

What?

Where?

Multi-Criteria Decision Analysis

Geo-Information Systems

MCDA: Wrap-up

- Huge field, often misused
- About the process, less about the outcome
- Highly participatory
- Often no right or wrong
- To make trade-offs and opinions explicit

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